ALEXANDRE VRAY

Robotics engineer (EPFL)

PROFILE

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avray02.github.io/Portfolio/

23 years old Ecublens, VD

LANGUAGES

French : Native

English : Professional

INTERESTS

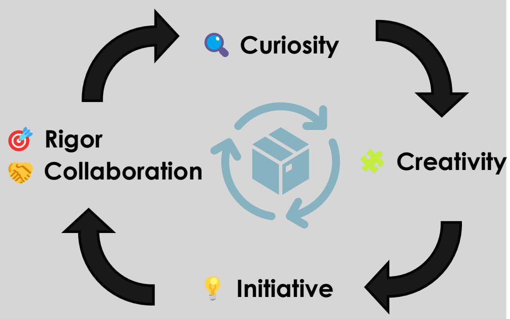
| Triathlon, climbing
Discipline, consistency, team spirit

| Mountains
Endurance & self-challenge

| Percussion (7 years)
Rhythm and coordination

| Personal projects
Technical creativity & tinkering (robotic arm)

PERSONAL VALUES



PROFILE SUMMARY

Graduated with a **master's degree in Robotic from EPFL**, I have **one year of R&D experience at Rolex SA,** combining my internship and master's project. I led highly technical projects focused on **analyzing complex components** and developing **real-time defect detection tools**.

Autonomous, rigorous, and results oriented, I adapt to demanding environments and **collaborate** effectively with multidisciplinary teams. My goal is to contribute to **innovative projects** combining technical analysis and product improvement, with tangible **industrial impact**.

EDUCATION

EPFL – Master's in Robotics (2022-2025) | 5.62/6
Specialization in mobile and industrial robotics

EPFL – Bachelor's in Microengineering (2019-2022) | 5.33/6
3rd year at Polytechnique Montréal | 3.8/4

Scientific Baccalaureate – Highest Honors (2019) | 17/20

PROFESSIONAL EXPERIENCE

R&D Project Engineer Intern | Rolex SA (2024 – 2025)
Designed an automatic defect detection system for watch components, developing the complete model and performing structured analysis of results in collaboration with production teams.

Traction Control Lead | EPFL racing team (2022-2023)
Developed a dynamic simulation and proposed an innovative method to estimate the vehicle's state in real time under strong constraints. Integrated and validated the solution in real-world conditions.

Teaching Assistant | EPFL (2021 –2024)
Provided supportive and adaptive guidance to students, addressing individual challenges and clearly conveying concepts through active listening and effective pedagogy.

KEY SKILLS

Proactive Problem-Solving
Designed defect detection tools at Rolex, introducing innovative solutions successfully validated in production.

Automation & Control
Development and optimization of control loops applied to complex systems.

Analytical Mindset & Results Focus
Analyzed complex data to guide actionable technical decisions

Collaborative Mindset
Provided support and active listening to facilitate teamwork.